

Welcome to UVM 2017 methodology

Verification at block and chip level needs structure, not only a DUT and a few directed tests

Why layered testbenches matter

- As designs grow, you need reusable agents, predictable phase order
- A flat procedural testbench can work for a tiny block, but it breaks when teams, protocols
- UVM gives you a shared vocabulary for environments, sequences, configuration

Two tracks, one idea

- Every lab module offers two ways to practice
- Makefile runs feel like work you will keep
- Agent intuition without installing UVM
- You may do either track, or both
- The usual rhythm is browser first for the idea, then offline runs for fidelity

Set up the real UVM track

- Obtain a UVM library and a simulator that can run it
- Open the legacy offline course tree linked from this repo
- Confirm you can compile and run a small testbench once
- Module twenty-two in this course is the dedicated offline run

Set up the browser lab track

UVM 2017 methodology (sketches)

learn_uvm2017 browser sketches for IEEE 1800.2-2017 UVM library (module tags). Fidelity remains offline UVM (module 22). TB anatomy / waves also under Waveforms.

Testbench layer diagram

test → env → agent + scoreboard → vif → DUT - 22 challenges - learn_uvm2017 01

UVM phase timeline

build → connect → run → check → report - 22 challenges - learn_uvm2017 04 - learn_pyuvm 05

Factory override sketch

types/init override - who gets constructed - 22 challenges - learn_uvm2017 05 - learn_pyuvm 06

ConfigDB key path

set/get by path + field - 22 challenges - learn_uvm2017 06 - learn_pyuvm 08

igital Design and Verification Platform

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ne Tools

Courses, labs, tools,...

seq_item / put / get / analysis - 22 challenges - learn_uvm2017 08

Sequence → driver flow

item → sequencer → driver → DUT - 22 challenges - learn_uvm2017 09 - learn_pyuvm 07

Objection raise/drop

who holds run open - raise/drop - 22 challenges - learn_uvm2017 10 - learn_pyuvm 08

Plusargs / CLP

+UVM_TESTNAME - seed/verbosity knobs - 22 challenges - learn_uvm2017 11

Scoreboard expect/actual

predict vs observe - orphans - 22 challenges - learn_uvm2017 13 - learn_pyuvm 09

Register model map

block→reg→field - front-door vs back-door - 22 challenges - learn_uvm2017 14

Multi-agent env

two+ agents - shared scoreboard - 22 challenges - learn_uvm2017 16 - learn_pyuvm 09

Virtual sequence

vssequencer refs - seq vs parallel - 22 challenges - learn_uvm2017 17

Callbacks sketch

pre/post hooks without subclassing - 22 challenges - learn_uvm2017 18

UVM reporting ladder

severity - verbosity - ID filter - 22 challenges - learn_uvm2017 19

Protocol checker sketch

bus rules vs scoreboard rules - 22 challenges - learn_uvm2017 20 - learn_api 09

VIP anatomy

agent + checker + coverage + docs - 22 challenges - learn_uvm2017 21 - learn_uart 12 - learn_tdc 09

UVM 2017 tools on the platform

How to move through modules

- For each module
- Python pyuvm is a separate course; this path stays with SystemVerilog UVM twenty seventeen
- When you finish this intro checklist, continue to the first lab module: testbench layers